

Exercise 2: Securing a Channel with TLS + Pre-Shared Keys

You should now be able to retrieve keys from the KMS. In this step, the goal is to use the previously retrieved key `f7e0edef-8bf4-43b2-983b-1e9d22ca6f13` to establish a TLS 1.2 PSK (pre-shared key) connection.

A HTTPS server using TLS PSK is running at `tii-qkd-workshop.uaenorth.cloudapp.azure.com:4443/`.

TASK: Write a TLS client (in a language of your choice) which can establish a TLS PSK connection with the server. Afterwards, over this channel, send a HTTP GET request to target `/result`. If everything works as expected, you should get a HTTP/1.1 200 OK, Content-Type: text/plain response - write down the content of the HTTPS body.

Hints

- **Hint 1:** Keep in mind that more commonly TLS is established with certificates. You therefore need to find a library that allows using PSK instead of certificates to establish the connection.
- **Hint 2:** Use the actual 44-character Base64 string as the key, not the decoded byte string.